Slenderness Ratio:

Given Data:

Column Dimentions:

b = 230 mm
d = 300 mm
L = 3050 mm

Beam Dimentions:

b = 230 mm
d = 450 mm
L = 6350 mm

Formulas,

$\mathbf{B_1, B_2 = \Sigma Kc/(\Sigma Kc + \Sigma Kb)}$

$\Sigma Kc = EIc/Lc$
$\Sigma Kb = EIb/Lb$
$I = (bd^3)/12$

Solution

For K major,

ΣKc,

Ic = 517.5x10⁶ mm⁴
Lc = 3050 mm
for ΣKc = 169672.1311 mm³

At the bottom joint, there are two column members contributing:

C7 : First → Second
← SELECTED MEMBER

First Floor

• ← BOTTOM JOINT of our selected member

C7 : Ground → First

So,

$$\sum K_c = K_{\text{Ground}\rightarrow\text{First}} + K_{\text{First}\rightarrow\text{Second}}$$

ΣKc = 2 x (EIc/Lc)
ΣKc = 339344.2623 mm³

ΣKb,

Ib = 1746.5625x10⁶ mm⁴
Lb = 6350 mm
for ΣKb = 275049.2126 mm³
B1, B2 = ΣKc/(ΣKc +ΣKb)
B1, B2 = 0.56

Given Data:

Column Dimentions:

b = 230 mm
d = 300 mm
L = 3050 mm

Beam Dimentions:

RB 8,
b = 300 mm
d = 450 mm
L = 3327 mm

RB 9,
b = 230 mm
d = 450 mm
L = 3264 mm

Formulas,

$$B_1, B_2 = \Sigma Kc/(\Sigma Kc + m\Sigma Kb)$$

$\Sigma Kc = EIc/Lc$
$\Sigma Kb = EIb/Lb$
$I = (db^3)/12$
$m = 1.5$

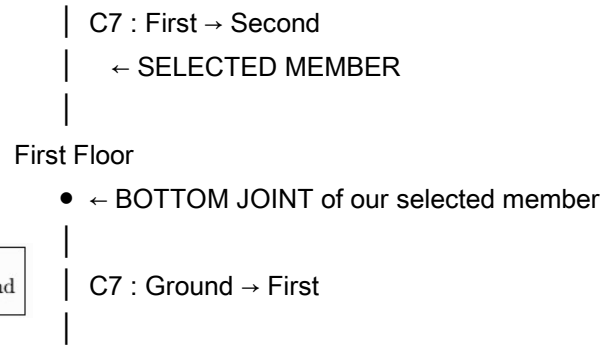
Solution

For K minor,

ΣKc,

Ic = 304.175x10⁶ mm⁴
Lc = 3050 mm
for ΣKc = 99729.5082 mm³

At the bottom joint, there are two column members contributing:



So,

$$\sum K_c = K_{\text{Ground}\rightarrow\text{First}} + K_{\text{First}\rightarrow\text{Second}}$$

ΣKc = 2 x (EIc/Lc)
ΣKc = 199459.0164 mm³

In unbraced frames, the beams are bent in double curvature. Therefore the beam stiffness in such frames should be modified as

$$K_b = 1.5 \times \frac{I}{L}$$

ΣKb, RB8

I_b = 2278.125x10⁶ mm⁴

L_b = 3327 mm

for ΣKb_1 = 684738.5032 mm³

ΣKb, RB9

I_b = 1746.5625x10⁶ mm⁴

L_b = 3264 mm

for ΣKb_2 = 535098.8051 mm³

B1 = :/(ΣKc +mΣKb1)

B2 Kc/(ΣKc +mΣKb2)

B1 = 0.1626157

B2 = 0.1990396

KL/D for Major Axis

K = 1.515

Lu = 3050 + 3050 -450

Lu = 5650 mm

In Etabs:

Lu = LF * L

Where,

LF = Length Factor

L = Over all length

length factor take it as,

LF = 1.852

Validation.

Lu = LF * L

LF = Lu/L

Lu = 5650 mm

L = 3050 mm

LF = 1.852459

* In the Major axis, There is no intermediate beam between first and second floor

KLu/D major

K	=	1.515	
Lu	=	5650	
D	=	300	* column Depth
Klu/D	=	28.5325	

KL/D Minor axis

K	=	1.2
Lu	=	3050 - 450
Lu	=	2600

In Etabs:

Lu = LF * L

Where,

LF	=	Length Factor
L	=	Over all length

length factor take it as,

LF = 0.852

Validation

Lu	=	LF * L	
LF	=	Lu/L	
Lu	=	2600	mm
L	=	3050	mm
LF	=	0.852459	

KLu/D major

K	=	1.2	
Lu	=	2600	
D	=	230	* column Width
Klu/D	=	13.565217	

Capacity Ratio

b	=	230	mm
D	=	300	mm
fck	=	25	MPa
fy	=	550	MPa
Pu	=	536.8438	kN
Mu_2	=	18.9251	kNm
Mu_3	=	52.2751	kNm

Asc = 18.9 cm² 1890 mm²

As n*(πd²/4)

Where,

n = No. of . Bars provided

d = Provided Dia. Of bar

n = 6 No's

d = 20 mm

As 1884.9556 mm²

Asc = As

Axial Capcity alone,

Gross Area, Ag = b x d

Ag = 69000 mm²

Concrete Area, Ac

Ac = Ag - As

Ac = 67115.044 mm²

Pu = 0.4 fck Ac + 0.67 fy Asc

Pu = 1367615.4 N

Pu capcity = 1367.62 kN

End Moment of the column obtained from Structural Analysis		Major Bending	Minor Bending
=	Mu_1	-10.7705 kNm	-0.3708 kNm
	Mu_2	5.0095 kNm	0.3954 kNm

IS 456, Clause 39.7,

Initail moment

initial moment equal to sum of 0.4 M_{u1} and 0.6 M_{u2}

Mi = 0.4 Mu1 + 0.6 Mu2

$$M_i + M_a \geq M_{u2}$$

for this Mi = -1.3025

0.4 Mu_2 = 2.0038

0.4 Mu2 > Mi, Not Safe

So, changing the Equation into

0.6 Mu1 + 0.4 Mu2

Will be Safer

$$M_i \geq 0.4M_{u2}$$

Initail moment for Major Axis

$M_{i\ major}$ = 0.6 Mu1 + 0.4 Mu2

0.6 Mu1 = -6.4623

0.4 Mu2 = 2.0038

$M_{i\ major}$ = -4.4585 kNm

Initail moment for Minor Axis

0.4 Mu2 = 0.15816 kNm

Major axial additional moment, M_a As per IS 456, Clause 39.7.1

$$M_a = \frac{P_u D}{2000}\left(\frac{L_e}{D}\right)^2$$

Pu = 536.8438 kN

D = 300 mm

K = 28.5325

(Pu*D)/2000 = 80.52657

K^2 = 814.10356

M_a = 65556.967

M_a = 65.56 kNm

Minor axial additional moment, M_a

$$M_a = \frac{P_u b}{2000}\left(\frac{L_e}{b}\right)^2$$

Pu = 536.8438 kN

b = 230 mm

K = 13.565217

(Pu*D)/2000 = 61.737037

K^2 = 184.01512

M_a = 11360.548

M_a = 11.37 kNm

As per IS456 clause 39.7.1.1

Applying Reduction factor K for Additional moment (K*Ma)

P_u = 536.84 kN

P_b = 142.8208 kN

P_{uz} = 1553.895 kN

$$k = \frac{P_{uz} - P_u}{P_{uz} - P_b} \leq 1$$

IS 456 Clause 39.6

Puz = 0.45 fck Ac + 0.75 fy Asc

fck = 25 MPa

Ac = 67115.044 mm^2

0.45 fck Ac = 755044.25

fy =	550	MPa
Asc =	1890	mm²
0.75 fy Asc =	779625	
0.45 fck Ac + 0.75 fy Asc =	1534669.2	
	1534.6692	kN

Pb $P_b \approx 0.45\left(k_1 + k_2\frac{p}{f_{ck}}\right) f_{ck} b D$

Given Data

b =	230	mm
d =	300	mm
d' =	58	mm
fck =	25	Mpa
fy =	550	Mpa
Ag =	69000	mm²
Asc =	1890	mm²

As per Sp 16, Table - 60

p = (Asc/ Ag) x 100

p = 2.74 %

Therefore,

P/fck = 0.1096

Calculate d'/D

d'/D = 0.1933333

$$y = y_1 + \frac{x - x_1}{x_2 - x_1}(y_2 - y_1)$$

k1,

0.15	0.196
0.193	0.18568
0.2	0.184

x = 0.193

x1 = 0.15 y1 = 0.196

x2 = 0.2 y2 = 0.184

K1 = 0.18568

For K_2, fy 550 not given in code book

so, K_2 = 0

Therefore,

Pb = 0.45* K_1* fck bD

= 142830 N

Pb = 142.83 kN

Reduction Factor, k = (P_{uz} - P_u)/(P_{uz} - P_b)

0.7207665

k*Ma for Major Axis

k = 0.7207665

M_a = 65.56

k*Ma = 47.26

k*Ma for Major Axis

k = 0.7207665

M_a = 11.37

k*Ma = 8.196

Minimum Ecentricity

Major Axis: e_{min} subjected to a minimum 20 (emin ≥ 20)

IS 456 Clause 25.4 states

e_{min} = (Lu/500) + (lateral dimension/30)

Lu = 5650 mm

D = 300 mm

e_{min} = 21.3

Since,

e_{min} = 21.3 > 20

e_{min} = 21.3 mm

Minor Axis

Lu = 2600 mm

D = 230 mm

e_{min} = 12.866667

Is456 requires,

emin ≥ 20

Therefore,

emin = 20 mm

MinimumMoment

M_{min} = P_u e_{min}

Major Axis

M_{min} = 11.434773 kNm

Minor Axis

M_{min} = 10.736876 kNm

Capacity Ratio

$$CR_{linear} = \frac{M_{u3}}{M_{3,cap}} + \frac{M_{u2}}{M_{2,cap}}$$

Mu3 = 52.2751
Mu2 = 18.9251

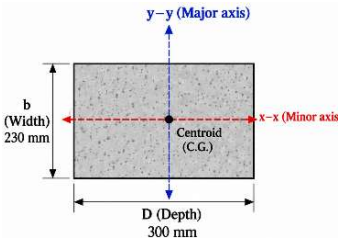
M3,cap = $$M_{3,cap} = M_{3,1} + \frac{P_1 - P_u}{P_1 - P_2}(M_{3,2} - M_{3,1})$$

Pu = 536.8438 kN
M3,1 = 66.7579 kNm
M3,2 = 77.8435 kNm
P1 = 627.9322 kN
P2 = 233.7011 kN

M3,cap = 63.319 kNm

For M2 cap

Assume, xu = 132.93 mm



Concrete Compression,

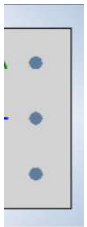
Cc = 0.36 fck bc xu
= 358911
Cc = 358.911 kN

compression Steel force,

Steel Strain εsc = 0.0035 ((xu - d')/xu)
d' = 58
xu = 132.93
εsc = 0.0019729

Steel Stress fsc = E * εsc
E = 200000
fsc = 394.57609 Mpa

Asc for compression side



n = 3
d = 20
Asc = n*(πd²/4)
= 942.4778

Therefore,

Cs = fsc*Asc
= 371.87921 kN

Tension steel force,

Distance of tension bars from compression face:

230 - 58 = 172 mm

Steel Strain $\varepsilon_{st} = 0.0035((d - x_u)/x_u)$

$d = 172$

$x_u = 132.93$

$\varepsilon_{st} = 0.0010287$

Steel Stress $f_{st} = E * \varepsilon_{sc}$

$E = 200000$

$f_{st} = 205.73986$ Mpa

Asc for compression side

$n = 3$

$d = 20$

$Asc = n*(\pi d^2/4)$

$= 942.4778$

Therefore,

$T_s = fsc*Asc$

$= 193.90525$ kN

Check axial equlibrium

$p = C_c + C_s - T_s$

$p = 536.9$ kN

Required, $Pu = 536.9$ kN

$Pu = p$ *equilibrium is satisfied.

Concrete Moment

Centroid of section = 230/2

$= 115$ mm

$= 0.42\ x_u$

$= 55.8306$ mm

Lever arm, Zc = 59.1694 mm

Therefore,

$M_c = 21236.549$

$= 21.236549$ kNm

Compression Steel Moment

Lever arm, Zs = 57

Therefore,

$M_{sc} = 21197.115$

$= 21.197115$ kNm

Tension Steel Moment

$$M_{st} = 11052.599$$
$$= 11.052599 \text{ kNm}$$

Major Capacity axis, $M_{2,cap}$

$$M_{2,cap} = M_c + M_{sc} + M_{st}$$
$$= 53.49 \text{ kNm}$$

Cpacity Ratio

$$CR = (M_{u3}/M_{3,cap}) + (M_{u2}/M_{2,cap})$$

$$(M_{u3}/M_{3,cap}) = 0.8255832$$
$$(Mu2/M2,cap) = 0.3538063$$

$$CR = 1.1793895$$